Muhammad Sharjeel

Electrical (Power) Engineering undergraduate

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Power systems, embedded control and electric mobility, developed through undergraduate design studies, practical prototypes and industry learning.

EDUCATION

Electrical (Power) Engineering — Air University Islamabad

September 2023–present

Undergraduate · In progress · Current reported CGPA 3.10/4.00 (Fall 2026 report snapshot).

Intermediate studies, Nishat College of Science for Boys, Multan, Pakistan · 2021–2023.

INTERNSHIP EXPERIENCE

Islamabad Electric Supply Company (IESCO)

July–September 2025

Engineering intern

A utility internship connecting power-engineering coursework with grid-station equipment, maintenance, protection and safe working practices.

Observed and assisted work associated with Substation and Transmission Lines (SS&TL) and Protection and Instrumentation (P&I).

POWERCHINA SEPCO1

July–September 2024

Engineering intern

An introduction to the coordination behind engineering delivery, with exposure to project management, procurement and planning.

Observed project-management and planning processes in an engineering organization.

SELECTED ACADEMIC PROJECTS

Design and Integration of Chassis and Power-train for a Range Extended Electric Vehicle

FYP-1 complete · Electric mobility

A two-seater academic REEV study connecting electric propulsion with an engine-generator range extender. FYP-1 established the architecture and integration methodology.

34 kW Micro Hydropower Plant Design

Completed design study · Power systems

A completed rural-electrification design study for Kewai Village, Naran, covering the water-to-wire system and its environmental and economic feasibility.

Autonomous Multi-Sensor Navigation Robot

Implemented semester prototype · Embedded systems

An implemented Arduino semester prototype combining line tracking, obstacle sensing and ambient-light response through embedded control.

Single-Phase Center-Tapped Transformer

Fabricated academic prototype · Power systems

An Electric Machines design-and-fabrication project connecting winding calculations with bobbin availability, core fit and practical redesign.

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Research, engineering skills & additional learning

RESEARCH AND WRITING

Strategic Policy Interventions for Enhancing the Future of Engineering Graduates in Pakistan

Published paper · 2025 · International Journal of Social Sciences Bulletin

Sobia Nasir Shafiq · Bushra Javed · Muhammad Sharjeel

A coauthored policy paper examining the relationship between engineering education, practical skills and graduate employment in Pakistan.

https://doi.org/10.5281/zenodo.16811464

Impact of Electric Vehicle Charging Infrastructure on Distribution Network Voltage Stability

Manuscript

Muhammad Sharjeel · Muhammad Zikyas Babar · Mashhood Un Nabi Butt

An analytical manuscript examining how EV charging demand affects voltage profiles, losses and stability margins on radial distribution feeders.

Design and Integration of Chassis and Power-train Structure for a Range Extended Electric Vehicle (REEV)

Accepted abstract · 2026

An abstract from the two-seater REEV academic project, accepted for a future oral presentation.

ENGINEERING SKILLS AND TOOLS

Power engineering

Power-system design · Transformer calculations and testing · Micro hydropower design · Substation fundamentals · Electrical safety

Embedded systems

Arduino UNO / ATmega328P · 8051 microcontrollers · Embedded C / C++ · Sensor integration · PWM and motor control · Proteus

Engineering software

MATLAB · C++ · Object-oriented programming · SolidWorks · PCB design

RELEVANT COURSE COMPLETION

Renewable Energy Technology Fundamentals

University of Colorado Boulder via Coursera · 2 April 2025

Electric Industry Operations and Markets

Duke University via Coursera · 26 March 2025

Essentials of Electrical Safety

Alison · 6 August 2025

NFPA 70E (2024) - Navigating Workplace Electrical Safety & Essential Updates

Alison · 11 August 2025

ACADEMIC AND ORGANISATIONAL RECOGNITION

Air University departmental Certificate of Commendation — Spring 2025

Air University departmental Certificate of Commendation — Fall 2025

IEEE Air University Certificate of Recognition, 2025–26

Certificate of Appreciation for organising Farewell 2026, ECE department

Project outcomes are academic work. REEV is at FYP-1 complete; hydro generation and payback figures are design estimates. Course certificates record completion and do not confer professional licensure. Full project and credential context: engr-sharjeel.github.io.